

The Life Cycle of Stars

Stars are born, live, and die over timescales of millions to trillions of years. A star's entire life is a balance between two opposing forces: the inward pull of gravity and the outward push of radiation pressure from nuclear fusion in its core. The mass of a star at birth is the single most important factor determining its life cycle and ultimate fate.

Star Formation

Stars form inside vast clouds of gas and dust called nebulae. When a region of a nebula becomes dense enough, gravity causes it to collapse into a hot, spinning core called a protostar. As the core contracts, its temperature and pressure rise. When the core reaches about 10 million kelvin, nuclear fusion of hydrogen into helium ignites, and a true star is born.

The Main Sequence

For most of its life, a star is on the main sequence, steadily fusing hydrogen into helium in its core. During this phase the star is in hydrostatic equilibrium: the outward radiation pressure from fusion exactly balances the inward force of gravity. The main sequence is the longest stage of a star's life; our Sun, for example, will spend roughly 10 billion years on it. More massive stars burn their fuel far faster and have much shorter lives.

The Fate of Low-Mass Stars

When a low-mass star like the Sun exhausts the hydrogen in its core, the core contracts and the outer layers expand and cool, turning the star into a red giant. Eventually the outer layers are gently expelled into space, forming a glowing shell called a planetary nebula. What remains is a small, dense, hot core called a white dwarf, which slowly cools over billions of years.

The Fate of High-Mass Stars

Stars more than about eight times the mass of the Sun follow a more violent path. They swell into red supergiants and fuse progressively heavier elements, building up to iron in their cores. Because fusing iron consumes rather than releases energy, the core collapses catastrophically and the star explodes as a supernova. The collapsed remnant becomes either a neutron star or, if the original star was massive enough, a black hole. The maximum mass of a stable white dwarf, about 1.4 times the mass of the Sun, is known as the Chandrasekhar limit.